

Technical Case Study: Cyber Attack Analysis

Student Name: Iftach Samel

Date: 14/6/2026

Chosen Cyber Technique: Man In The Middle - Evil Twin

Selected Cyber Attack/Campaign: Australian airport Evil Twin attack

Case Overview

Attribute	Details
Attacker/Threat Actor	A 42 year old man from Palmyra/Australia (his personal information is classified)
Victim/Target Organization	Targeted primarily individual airline passengers, affected multiple airports and aircrafts.
Attack Timeline	<u>April 2024</u> - Airline staff detect a suspicious Wi-Fi network mid flight. <u>19 April, 2024</u> - The AFP checks the man's luggage in Perth, and finds his devices. <u>8 May 2024</u> - The suspect is arrested and charged with multiple cybercrime allegations related to access to people's personal data and faking public Wi-Fi networks.
Overall Impact & Damage	The attacker harvested large amounts of personal information from victims, including login details and online data.

Technical Attack Architecture & Technique Execution

- Initial Access Vector: The attacker gained access by setting public APs in high-traffic public areas (airports and airplanes on domestic flights). The attacker mirrored legitimate Wi-Fi network identifiers, which led unsuspecting individuals to give their data.
- Mechanics of the Chosen Technique: The attacker exploited the 802.11 Wireless protocol, which lacks a handshake and makes infiltrating into public networks easy. In addition, the attacker used DNS/DHCP redirection which redirected connected users into an imposter login page which harvested their data. No commands or malicious payload was transferred.

- Attack Flowchart/Timeline:
 - Step 1 - The attacker deploys a portable wireless access device that mimics a public AP and creates the rogue AP.
 - Step 2 - The victims connect to the rogue AP, which gives them their IP addresses and intercepts their web traffic.
 - Step 3 - The rogue AP gives the victim a fake authentication page which asks them to give their personal information and gets it instead.

Detection Analysis

- Historical Detection: The detection was by the suspecting airline crew (a third party), that noticed another public network that tries to mimic a legitimate one. The attack started and went undetected for weeks and even months.
- Proactive Detection: There could have been tracking of the legitimate public networks' SSID and MAC addresses, which can help distinguish between normal APs and rogue ones. There could have also been sniffing of DHCP and DNS packets to see suspicious behaviour, and overall the airports and airlines should have observed for such attacks.

Key Takeaways & Mitigation

- Using VPN for all public APs can assure that the network can't abuse the access point to gain unwanted information.
- Moving from 802.11 protocol to a more trusted one.
- Tracking the public networks can reduce rogue APs.
- Lessons For Intrusion Detection: Current ways to detect rogue APs rely on humans, there needs to be a standardized model (even a weak one) that will detect anomalies over the network.

References

- [Reference 1: Australian Federal Police, "Man charged over creation of 'evil twin' free WiFi networks to access personal data,"
<https://www.afp.gov.au/news-centre/media-release/man-charged-over-creation-evil-twin-free-wifi-networks-access-personal>]
- [Reference 2: MITRE ATT&CK Technique T1557 (Adversary-in-the-Middle),
<https://attack.mitre.org/techniques/T1557/>]
- [Reference 3: Preprints.org, "Evil Twin Attack in Wi-Fi Networks: Evolution, Mutation Taxonomy, and Exposure Time Analysis (2005–2026)," <https://www.preprints.org/manuscript/202606.0325>]